

SKINNER'S SCIENTIFIC SPECTRUM

“As a race of intelligent beings, humans have thus far prospered incredibly well on the journey to uncover the hot boiler room of the universe – the ‘big bang.’

- NEO SKINNER

MATHEMATICS

COMPLEX NUMBERS, PLANE AND SIMPLE

In your maths class, you will invariably have been told that you cannot find the square root of a negative number, and whilst you scratch your head pondering the solutions to a quadratic equation with a discriminant less than zero ($\Delta < 0$) your maths teacher has screamed ahead with the content leaving you in the dust of non-intersecting hyperbola. But, let us now take a step back and consider where the solutions may lie to that puzzling equation and heed not the advice of your wizened maths teacher!

Quadratic equations of the form $ax^2 + bx + c = 0$ can be solved to find values of x that make the overall equation equal to zero. We determine the number of roots using a function called the discriminant, Δ , which consists of the radicand of the quadratic formula - the part under the square root:

$$\Delta = b^2 - 4ac$$

The Quadratic Discriminant

When the discriminant is less than zero, the quadratic equation has no real roots as the square root of a negative number is usually said to be undefined because with the square of any real number, it is only ever possible to produce positive real numbers.

$$\mathbb{R}^2 = \mathbb{R}^+$$

The Square of Any Real Number

However, we can still imagine a number with the opposite property, where the square of our imaginary number produces a negative result. More specifically, let us imagine a number (i) that gives us negative one when squared:

$$i^2 = -1$$

The Square of an Imaginary Number

Using this definition, we can consider the solutions to a quadratic equation with a negative discriminant by looking at the quadratic formula:

$$\frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

The Quadratic Formula

The section under the square root - the radicand - is the familiar discriminant from earlier, and if we make this value negative and use our imaginary number to evaluate the full formula, we find that a number of two parts can be calculated, one part real and one part a multiple of our imaginary number, which we call a complex number (z):

$$z = a \pm bi$$

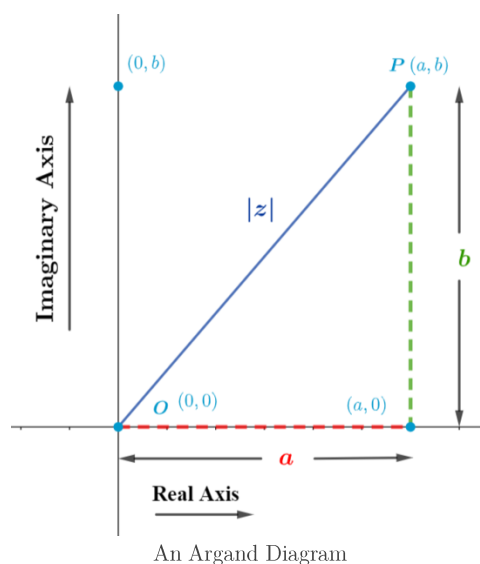
A Complex Number

The real part comes from the section outside the square root and the imaginary part comes from the radicand inside the square root. Neat, huh?

We have now developed a method for solving a quadratic equation with a negative discriminant and can find two solutions for this case (from the $\pm$ sign) which are both complex and vary only by the sign between the real and imaginary parts. We call this pair of solutions a complex conjugate.

It is worth mentioning here that our imaginary number does not affect the final result of the calculation as it still solves our quadratic equation to be equal to zero, only that we have defined a solution outside of the real numbers. This has some pretty big implications, with the largest being how to represent them graphically.

To represent a complex number graphically, we have to define a new plane, alongside the real plane ($\mathcal{Re}$) we now also have an imaginary plane ($\mathcal{Im}$) and can plot a complex number as a real part on the x -axis and an imaginary part on the y -axis. We call this graphical representation an Argand diagram.



So, now you know! We can find the roots of a quadratic equation with a negative discriminant; we just need to be careful and define an entirely new plane for mathematics. But, your maths teacher is still not wrong, there *really* are no *real* solutions.

PHYSICS

MAXWELL'S EQUATIONS THROUGH THE MEDIUM OF HEALTH AND SAFETY

A ground fault circuit interrupt (GFCI) device may sound like a fairly mundane and wordy item that you can purchase at your local ironmonger. But, in fact, GFCI devices save millions of lives every year and are an essential piece of equipment for every building in the United Kingdom. An electrician will tell you that it 'trips a breaker' when live current flows through ground, but how?

To realise why, we need to uncover the secrets of Maxwell's equations which describe the electromagnetic force responsible for the electricity flowing around your home today. Maxwell wrote of four equations to explain the electromagnetic field and today we will focus on the most interesting of them – Faraday's Law of Induction.

In the cellar of the Royal Institution around 1820, Michael Faraday was playing around with a new thing called electricity and discovered that a current in a wire created a magnetic field. This phenomenon is called electromagnetic induction and can be found in everything from electric motors to induction hob ovens. Although Faraday did not discover how this worked, luckily James Clerk Maxwell swooped in to finish the job and produced the law of induction as the third in his series of four equations describing electromagnetism:

$$\nabla \times \mathbf{E} = -\frac{\partial \mathbf{B}}{\partial t}$$

Maxwell's Third Equation of Electromagnetism, The Maxwell-Faraday Equation

Do not worry about the maths, this law basically states that for an electric current flowing in a closed loop with an associated electric field, there will be an induced magnetic field flowing around the wire.

How, then, does this explain our neat GFCI devices? In a plug socket, there are three cables; live and neutral create a circuit, with current flowing from live to neutral, and ground is there in case anything goes wrong, in which case current will 'leak' back through the ground cable instead of neutral.

In normal operation, the current flowing through the live and neutral cables will be equal and thus the electric fields surrounding the cables will be equal and opposite, cancelling each other and leading to no magnetic field being induced.

However, if current flows through the ground cable or leaks elsewhere from a faulty appliance then there is a difference in current flow between the live and neutral cables and an uneven electric field is produced around the two cables. This, in turn, induces a magnetic field around the circuit.

A GFCI device functions by wrapping a coil of wire around the two cables in the socket which will gain an induced current when a magnetic field induces an electric field around it. Under

normal conditions, there is no magnetic field to induce a current in the coil, so current will only flow if there is a fault with the circuit. A-ha! We now have a way of detecting a fault.

When a fault occurs, the current flowing through the coil runs through a solenoid, which creates a magnetic field to move a metal contact away from its position, breaking the circuit such that current can no longer flow. The GFCI device has successfully protected you against the faulty appliance.

With the example of GFCI behind us, let us now delve into the mathematical belly of the beast, but do not fear! We will use simple explanations of the complicated maths involved for your enjoyment.

$$\nabla \times \mathbf{E} = -\frac{\partial \mathbf{B}}{\partial t}$$

Maxwell's Third Equation of Electromagnetism, The Maxwell-Faraday Equation

Starting from the left of the equation, we have the curl ($\nabla \times$) of the electric field ($\mathbf{E}$). The curl operator can be thought of as how much a field is rotating around a point, in this case, the electric field. In our above example of normal operation in a GFCI circuit, the cables form a closed loop, so the rotation clockwise cancels the rotation counterclockwise and thus there is no curl of the electric field ($\nabla \times \mathbf{E} = 0$).

What we actually mean here by the electric field, $\mathbf{E}$, is the electric field strength or what force a point charge of +1 C (Coulomb) would experience at a given point in the field. So, the value of $\mathbf{E}$ at a point x if a point charge of +1 C experiences a force of 1 N (Newton) is simply 1 ($\mathbf{E} = 1 \text{ N C}^{-1}$) measured in units of Newtons per Coulomb.

On the right of the equation, we have the negative partial derivative[†] ($-\frac{\partial f(x)}{\partial t}$) of the magnetic flux density ($\mathbf{B}$) with respect to time (t). This can be expressed more simply as the rate of change of magnetic flux with time and simply means how much magnetic flux flows into and out of a point per unit time (second).

The magnetic flux density, $\mathbf{B}$, can be directly derived from the magnetic field strength ($\mathbf{H}$) by the permeability (μ_m) of the medium the field passes through, where the magnetic field strength is defined similarly to the electric field strength and the permeability is just how easily a magnetic field passes through a medium.

$$\mathbf{B} = \mu_m \mathbf{H}$$

The Permeability Equation

If we then put it all together, Maxwell's third equation tells us that a moving (rotating) electric field produces a change in magnetic flux and hence inducing a magnetic field where there was not previously one.

ARTICLE BY NSR

[†] You need not worry about these; they act very similarly to normal derivatives with only a few minor differences.



CHEMISTRY

WHY IS THAT THE SHAPE OF ETHANOL? (VSEPR THEORY EXPLAINED)

My chemistry teacher was on the surface of Mars sometime around 1984 walking his space dog. ‘Space dog?’ you say, ‘What’s one of those - and what was your chemistry teacher doing on Mars?’ Well, thank you for shattering the dramatization - on with the anecdote. My chemistry teacher calls ethanol the ‘space dog’ as the structure of a molecule is vaguely like a dog, but strange enough that it could be some other-worldly being. So where does this come from? And how can we predict the shapes of molecules?

VSEPR or valence shell electron repulsion theory is a really fancy name for what is essentially predicting how much an electron in an atom will repel other electrons in neighbouring atoms. We then use a bunch of these predictions to form the shape of a molecule. Electrons in an atom arrange themselves to minimise the effect of repulsion and, when electrons form a bond, where atoms in a molecule are located is a simple product of this repulsion rearrangement.

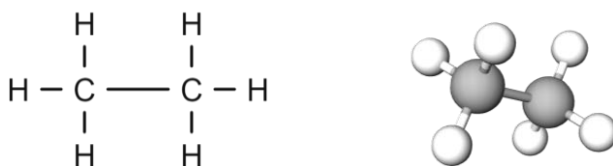
Methane - CH_4 - is one of the simplest molecules to study due to its remarkably tidy bonding. In essence, this molecule consists of a central carbon atom surrounded by four hydrogen atoms held in place by strong covalent bonds. But, what of its shape? Carbon has four valence (outer shell) electrons, and these are all bound with the single electrons of hydrogen, forming four pairs with one hydrogen atom per electron. All our electrons are neatly bound with no surprises; therefore the question becomes: how can we minimise the effect of the mutual and - crucially - equal repulsions in three dimensions?

In three dimensions, the four electrons (and, by extension, hydrogen atoms) form the vertices of a regular solid since all of the electrons must be held at the same angle to each other by their equal mutual repulsions. The 3D shape with equal internal angles and four vertices is called a tetrahedron and has central (bond) angles of 109.5° , this is the shape of methane!



The structure of methane; LTR: structural formula of methane, tetrahedron shape, ball-and-stick structure of methane

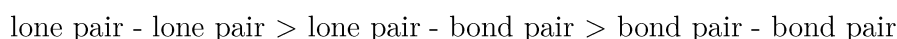
Let us keep building up the layers and look at the next closest molecule to ethanol, ethane - C_2H_6 . This time, each carbon atom is bonded to three hydrogen atoms and another carbon atom, but, since all the electrons are bonded it is still kept neat and it may be relatively simple to guess the shape of the structure of ethane, two tetrahedrons arranged to touch at one vertex.



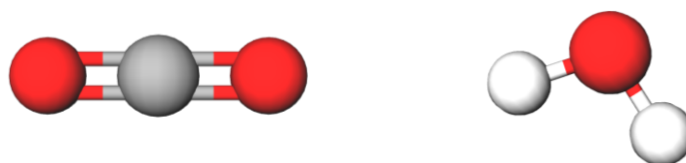
The structure of ethane; LTR: structural formula of ethane, ball-and-stick structure of ethane

Ok, now onto the big-ticket item. Ethanol - $\text{C}_2\text{H}_5\text{OH}$ - is the key ingredient of alcohol, and with the average Brit drinking 9.9 litres (L) of alcohol per year¹, it is a very important molecule. But, what does it look like? Ethanol introduces an additional oxygen atom to the structure of ethane, and this makes all the difference, as our electron bonds are no longer nicely balanced as oxygen has six electrons to loan to the effort, of which only two are bound to one carbon and one hydrogen atom. The other four electrons become a nuisance in our molecule and form two of what are known as lone pairs, a pairing of two unbound electrons in the valence (outer) shell of an atom.

VSEPR theory states that two lone pairs have a greater mutual repulsion than a bond and a lone pair and two bonds respectively, with the order of repulsion power as follows:

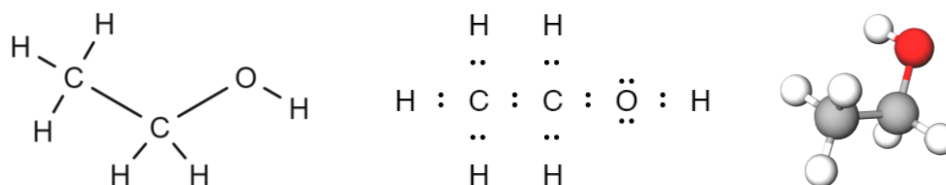


This means that the repulsion of the two lone pairs significantly affects the angle of the bonds around the additional oxygen atom. Normally, molecular geometries of three atoms, such as carbon dioxide - CO_2 , have a linear structure with all atoms falling along a line with bond angles of 180° , however, when lone pairs are introduced the linear structure is bent first to a bond angle of 119° (with one lone pair) and then further to 104.5° (with two lone pairs) in molecules such as water - H_2O . You need not worry about where the pattern lies with these repulsions, just know that they occur and affect the structure of our molecules.



The structures of three-atom molecules; LTR: ball-and-stick structure of carbon dioxide, ball-and-stick structure of water

If we put all of this together, we then get the structure of ethanol - a space dog!



The space dog; LTR: structural formula of ethanol, Lewis structure of ethanol, ball-and-stick structure of ethanol

Ethanol is comprised of the structure of ethane, with two tetrahedra joined at a vertex, joined to that of water, with a bent structure to form a fantastic fusion of VSEPR theory.

So, next time you are walking your space dog on the surface of Mars, just think of the four extra pesky electrons that screwed up the structure enough to form the very dog you are walking.

¹ The British Beer & Pub Association (2023). *Statistical Handbook 2023*.

RECOMMENDATIONS

A-LEVEL WIDER INDEPENDENT STUDY

- 📖 **Mathematics:** Research Euler's Formula for Polyhedra.
- 📖 **Physics:** What Is the Higgs Boson?
- 📖 **Chemistry:** How Do Artificial Muscles Work?

DISCRETE FURTHER READING

- 📖 **Mathematics:** [Chalkdust Magazine Issue 18, The Josephus Problem](#) [↗](#)
- 📖 **Physics:** [APS Physics Magazine Issue 17, Shedding Light on the Thorium-229 Nuclear Clock Isomer](#) [↗](#)
- 📖 **Chemistry:** [Chemistry World Magazine April 2024, Where Does Bromine Come From?](#) [↗](#)

CONTINUOUS FURTHER STUDY TOPICS

- 📖 **Mathematics:** The Modulus-Argument Form
- 📖 **Physics:** Gauss' Law
- 📖 **Chemistry:** The AXE Method

MONTHLY PROBLEM

Solve for x :

$$\frac{x-6}{2022} + \frac{x-5}{2023} + \frac{x-4}{2024} = 3$$

PROBLEM BY IMO, SUGGESTIONS BY NSR

CREDITS

SOURCES & PROVENANCE

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- 📖 Attwood, G., Berry, D., Bettison, I., Cope, L. and Garry, T. (2018). Edexcel AS and A level Further Mathematics Further Pure Mathematics 2 Textbook. Harlow, United Kingdom: Pearson Education Limited.

Physics:

- 📖 www.maxwells-equations.com. (n.d.). Maxwell's Equations. [online] Available [here](#) [↗](#).

Chemistry:

- 📖 Bhuyan, S. (2023). VSEPR Theory: Explanation, Chart, and Examples. [online] Chemistry Learner. Available [here](#) [↗](#).

- Haas, K. (2019). 3.2.1: Lone Pair Repulsion. [online] Chemistry LibreTexts. Available at: [here](#) .

Images:

- SG0039 (2021). Argand Diagram. [Digital] Available [here](#) .
- Tomruen (2006). Image d'un tétraèdre. [Digital] Available [here](#) .

Equations:

- Equations by [L^AT_EX](#) .
- Equations modified for publishing by [Lagrida](#) .

Chemical Diagrams:

- Chemical models by [MolView](#) .
- Lewis structures by [ChemDoodle](#) .

Formatting & Typesetting:

- Formatted in Microsoft Word
- Spellchecking by **grammarly**
- Body typeset in CMU Serif
- Title typeset in **TRADE GOTHIC INLINE**

Data sources are cited within the parent article.

EDITORS

- NSR:** Neo Skinner
- Your name could go here! (see below)

ACKNOWLEDGEMENTS

The editor would like to extend their gratitude to Mr A. Browning for inspiring over 40 years of students within the discipline of chemistry. Your contributions will remain unmatched for as long as the space dog continues to bark.

APPEAL

The editor is looking for a biology writer willing to put time and effort into producing a monthly article and recommendations for this publication. Hopeful applicants and deranged scientists should email their particulars to the editor, Neo Skinner, at skinner-science-spectrum@magna-aspirations.org.